

Three-group paired study: complete exploratory results

Souray Meng | 05 September 2026 | Native CFP and OCT task networks connected through R&B

129

accepted stage-two results

3

training seeds

3 x 3 x 4

groups x widths x mechanisms

1. Fixed uncentered SVD remains a strong reference. It reaches 68.83-69.47% mean F1 across the three widths; the smallest width has the lowest observed seed SD among its configurations.
2. Centered fitting does not reproduce the first-seed gain consistently. Wider configurations have slightly higher means, but substantially larger seed variation.
3. Learning the channel encoder and decoder is feasible, but not a demonstrated improvement. Mean F1 rises with width and remains below uncentered SVD at every tested width.

One comparison standard, three explicit groups

108 main results plus 21 reference results; shared baselines are counted once.

Group	Channel transform	Mechanisms	Seeds x widths	Results
U: uncentered SVD	Fixed Q from E[xxT]	Radon / self / scrambled / resample	3 x 3	36
C: centered-fit SVD	Fixed Q from E[xxT] - mean meanT	Same four mechanisms	3 x 3	36
L: learned channel	Learned encoder + decoder	Same four mechanisms	3 x 3	36
References	No bridge / learned CM / random QR	Shared baseline / standard Radon	3 + 9 + 9	21

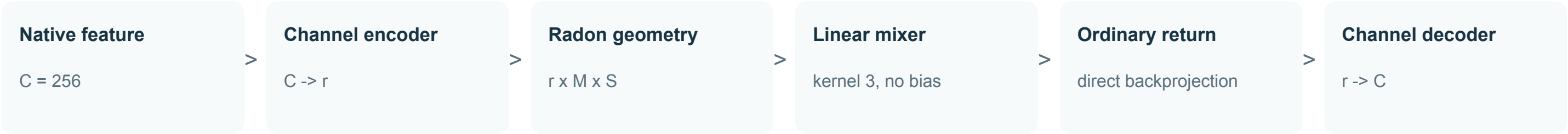
Fixed: stage3, M32, S64; rho = 1/16, 1/8, 1/4 gives r = 16, 32, 64 and per-source h = 512, 1024, 2048. Backbone LR 6e-5; head/bridge LR 1e-4. Batch 16, AdamW, weight decay 0.01; separate branch/bridge gradient clipping at 5.

Training: full native-network updates and BN updates. Shared independently trained parent checkpoints within each seed; same data order. At least 8 epochs; plateau after 6 epochs without improvement >0.001; LR x0.3 after 3 stagnant epochs. The 60-epoch cap is not a completion criterion.

Selection: one common stage-two checkpoint by mean of CFP/OCT macro-F1. All 129 results reached the prespecified plateau.

A shared geometric bridge, different channel maps

All bridge paths are linear at fixed parameters; the native task networks remain nonlinear and trainable.



Decoder output is a residual, added to the original native feature at its original MHD Node ID. Each branch retains its own backbone, head, cross-entropy loss and predictions.

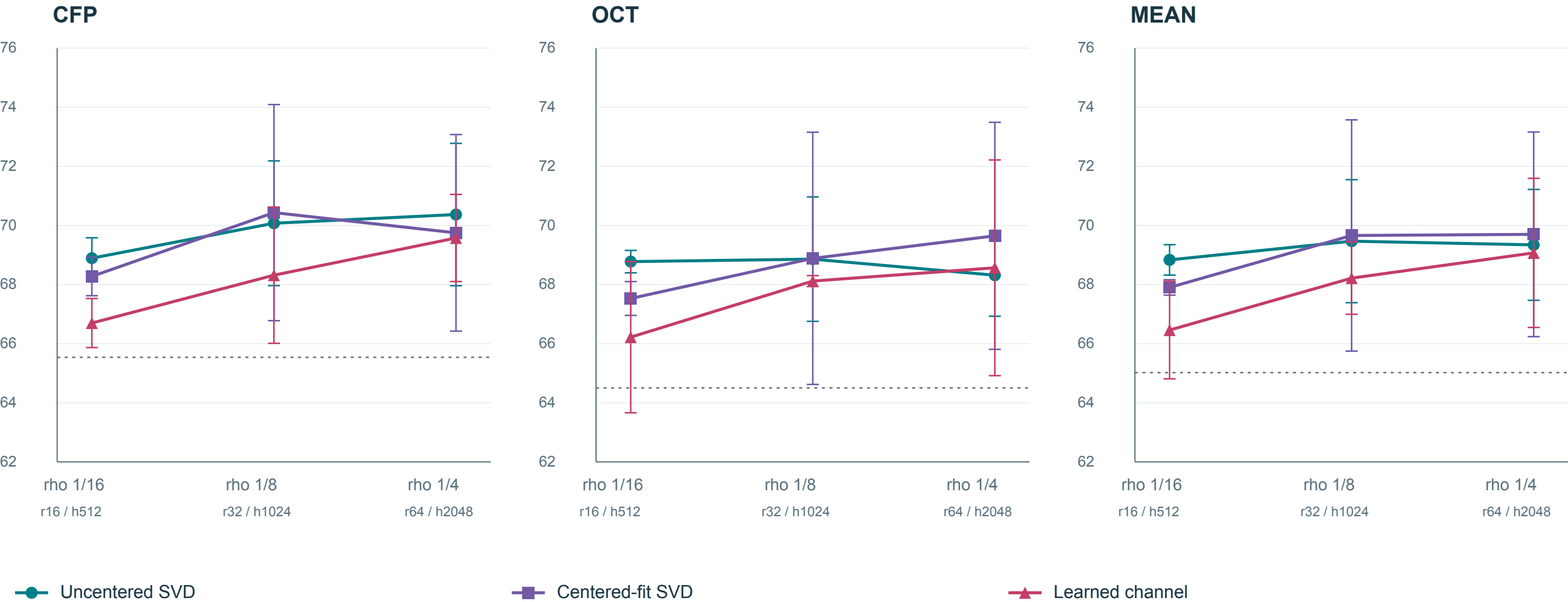
Group	Encoder / decoder	Trainable bridge modules
U	Q^T / Q ; fixed uncentered basis	Only the middle bridge convolution
C	Q^T / Q ; fixed centered-fit basis	Only the middle bridge convolution
L	E / D ; initialize Q^T / Q from QR	Encoder, middle convolution and decoder

Important distinction: C subtracts the training channel mean only when fitting Q. Its runtime never subtracts or adds that mean. L starts from the same QR directions as the random reference, but E and D train independently; they need not remain orthogonal or transposes.

Learned CM is a separate legacy reference: it compresses the flattened C x M channels after Radon.

Main comparison: performance across communication widths

Macro-F1 (%): three-seed mean +/- sample SD. Error bars show seed variation; dotted lines show no bridge.



A narrow peak is not required for validity. Uncentered SVD is comparatively flat across rho; learned-channel performance increases over the tested range. Neither pattern establishes a theoretical ceiling or a universal monotonic law.

Main numerical results: branches and baseline gains

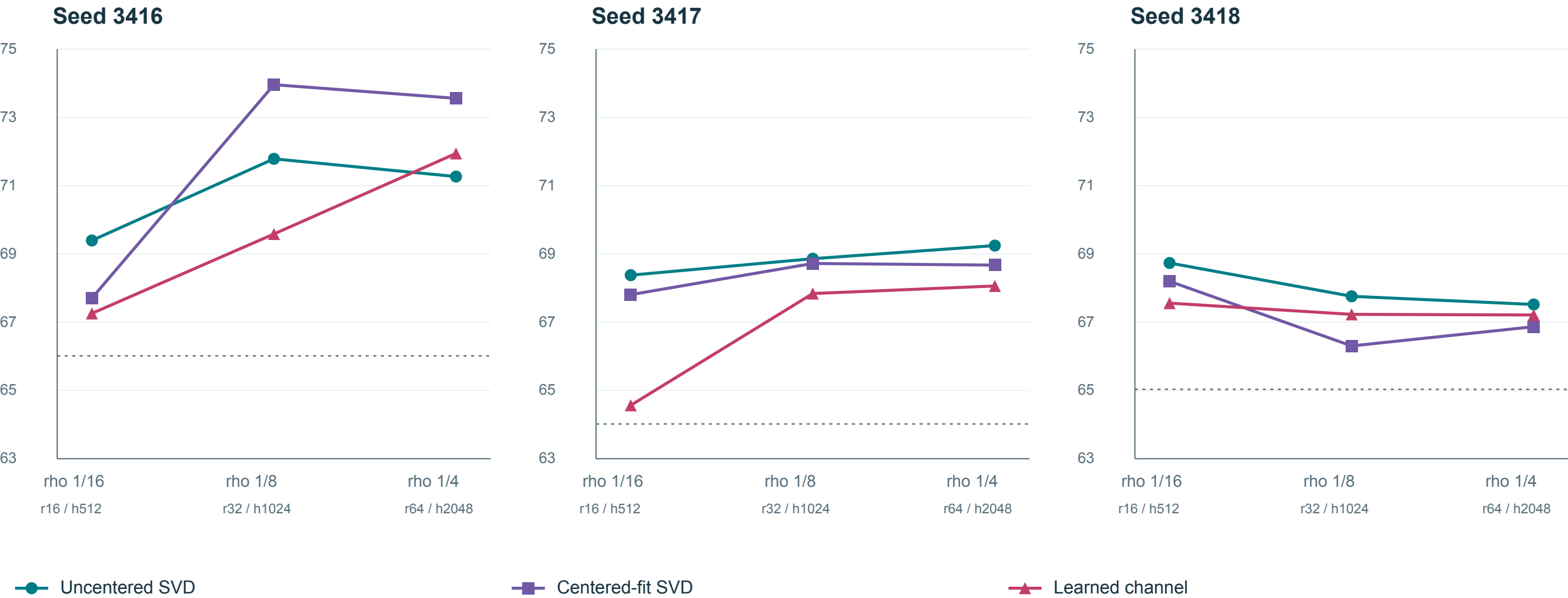
F1 in percent; +/- is the sample SD across three seeds. Baseline mean F1: 65.02%.

rho	Group	CFP	OCT	Mean	Gain (pp)
1/16	Uncentered SVD	68.89 +/- 0.69	68.78 +/- 0.38	68.83 +/- 0.52	+3.81
1/16	Centered-fit SVD	68.28 +/- 0.65	67.53 +/- 0.57	67.90 +/- 0.26	+2.88
1/16	Learned channel	66.70 +/- 0.83	66.22 +/- 2.56	66.46 +/- 1.65	+1.44
1/8	Uncentered SVD	70.08 +/- 2.11	68.86 +/- 2.10	69.47 +/- 2.08	+4.45
1/8	Centered-fit SVD	70.43 +/- 3.66	68.89 +/- 4.27	69.66 +/- 3.91	+4.64
1/8	Learned channel	68.32 +/- 2.30	68.12 +/- 0.19	68.22 +/- 1.22	+3.20
1/4	Uncentered SVD	70.37 +/- 2.41	68.32 +/- 1.39	69.34 +/- 1.88	+4.32
1/4	Centered-fit SVD	69.75 +/- 3.33	69.65 +/- 3.84	69.70 +/- 3.46	+4.68
1/4	Learned channel	69.58 +/- 1.47	68.57 +/- 3.65	69.07 +/- 2.52	+4.05

Centered fitting has the highest descriptive mean at rho 1/8 and 1/4. That does not establish superiority: the advantage is small relative to seed variation and the paired intervals cross zero.

Replication: show each training seed separately

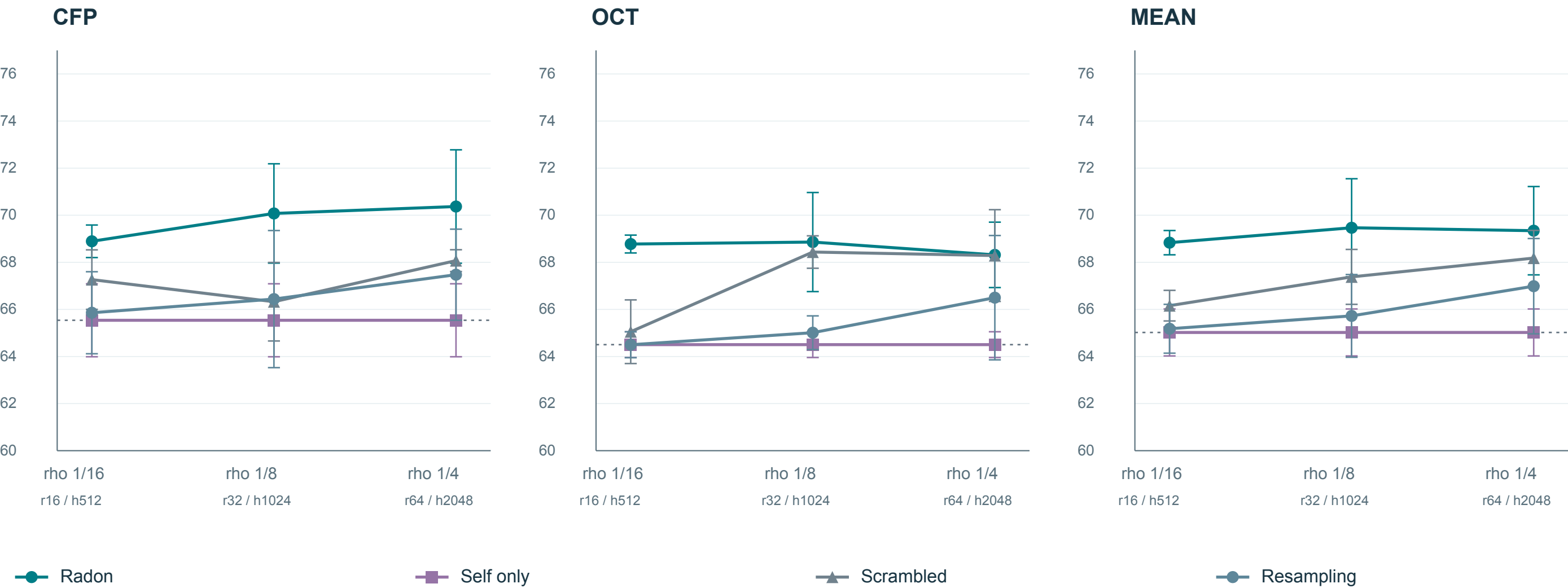
Mean macro-F1 (%); dotted lines show no bridge. Seeds 3416/3417 informed selection; 3418 is a training repeat, not independent test data.



The early centered-fit peak in seed 3416 does not recur at the same magnitude in 3417 or 3418. Seed-wise directions are retained in all paired tables.

Uncentered SVD: what does the communication mechanism add?

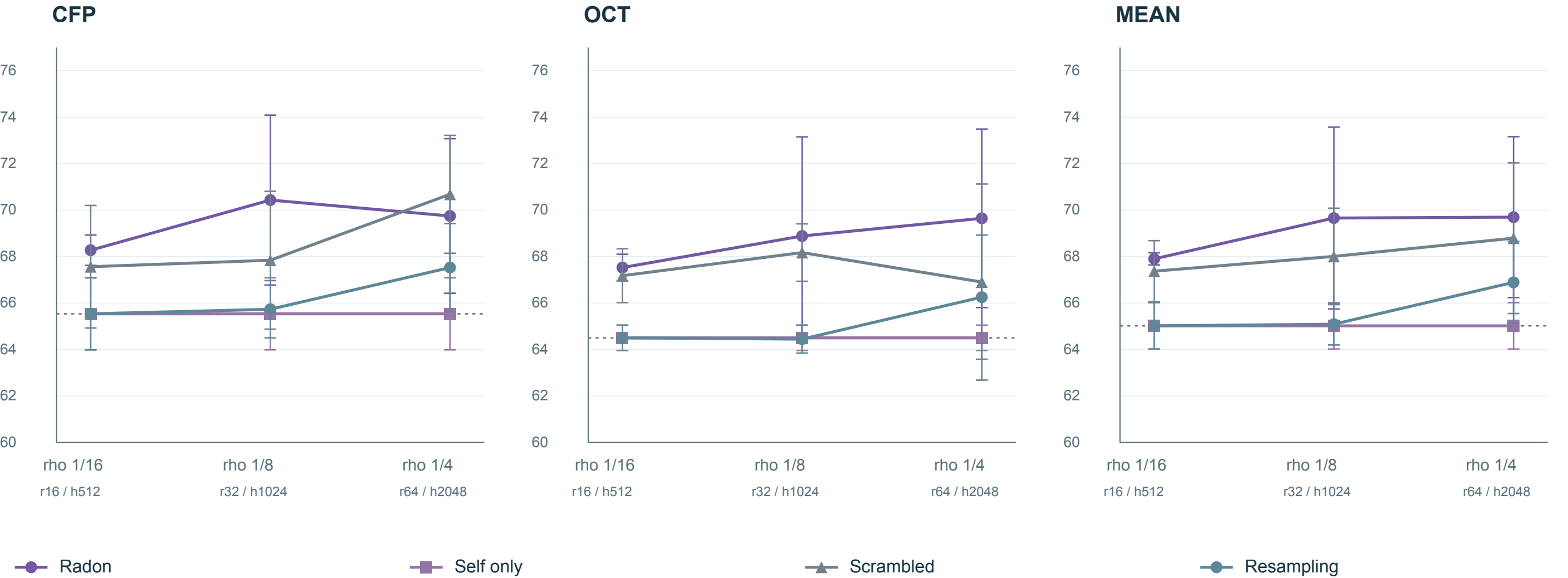
Macro-F1 (%): mean +/- seed SD; dotted lines show no bridge. Matched channel family, rho, parents and training.



Self-only uses a block-diagonal mixer mask and has fewer effective parameters. Scrambling preserves channel maps but changes spatial organization. Resampling matches width and operator row norms, not geometry or singular spectrum.

Centered-fit SVD: what does the communication mechanism add?

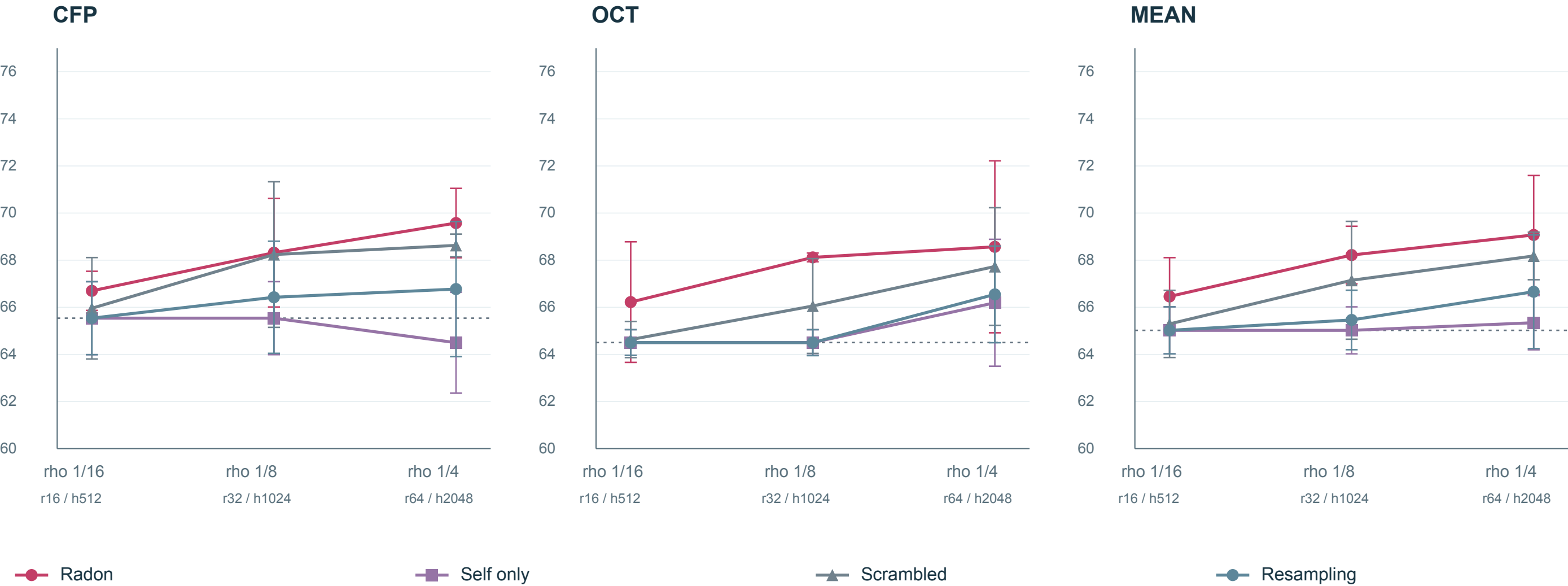
Macro-F1 (%): mean +/- seed SD; dotted lines show no bridge. Matched channel family, rho, parents and training.



Self-only uses a block-diagonal mixer mask and has fewer effective parameters. Scrambling preserves channel maps but changes spatial organization. Resampling matches width and operator row norms, not geometry or singular spectrum.

Learned channel: what does the communication mechanism add?

Macro-F1 (%): mean +/- seed SD; dotted lines show no bridge. Matched channel family, rho, parents and training.

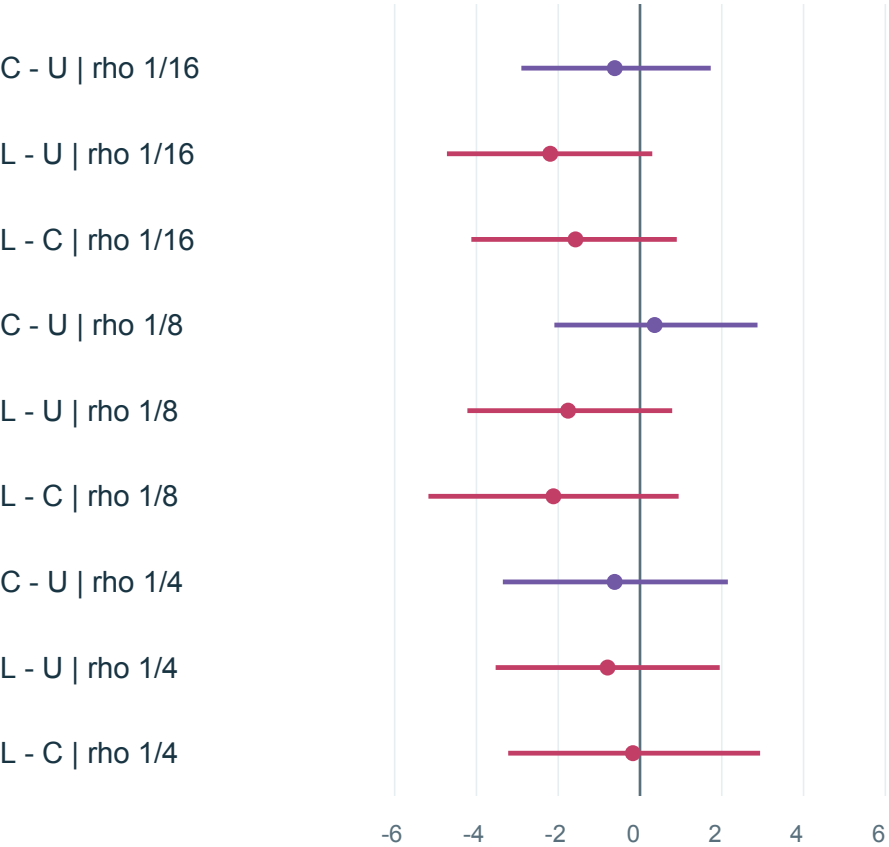


Self-only uses a block-diagonal mixer mask and has fewer effective parameters. Scrambling preserves channel maps but changes spatial organization. Resampling matches width and operator row norms, not geometry or singular spectrum.

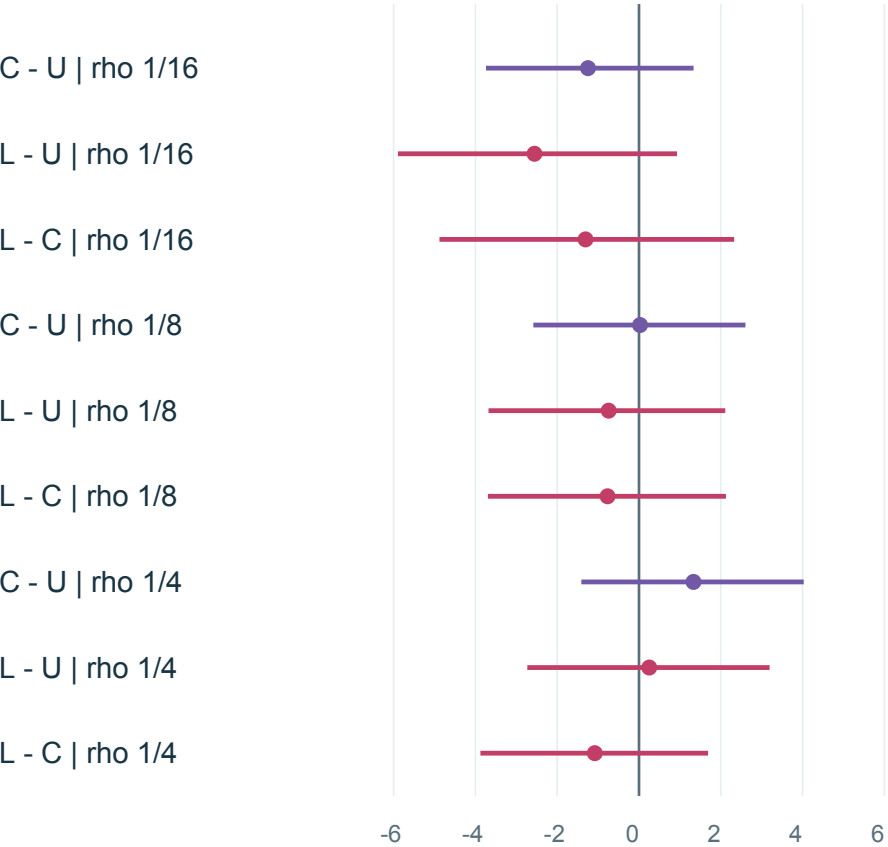
Matched group differences: uncertainty stays visible

F1 difference (percentage points), 95% paired intervals; 10,000 shared participant resamples across seeds. No multiplicity correction.

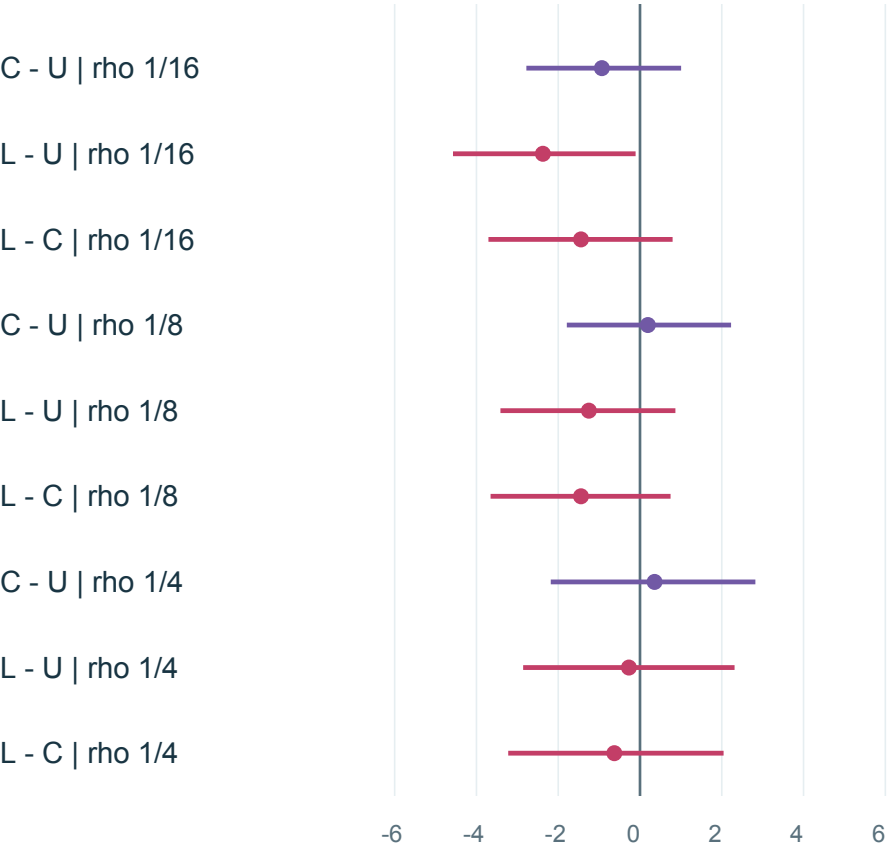
CFP



OCT



MEAN



Intervals condition on the trained models and their development-selected checkpoints. They do not include uncertainty from retraining, and do not turn the three copies of each participant into independent observations.

Additional references: width, performance and parameter cost

The three primary groups remain separate from learned CM, random QR and the shared no-bridge baseline.

rho	Method	Mean F1 +/- SD	Bridge params (M)	Total params (M)
1/16	Learned CM	68.27 +/- 1.94	19.923	64.262
1/16	Random QR	65.47 +/- 1.27	3.146	47.484
1/16	Uncentered SVD	68.83 +/- 0.52	3.146	47.484
1/16	Centered-fit SVD	67.90 +/- 0.26	3.146	47.484
1/16	Learned channel	66.46 +/- 1.65	3.162	47.501
1/8	Learned CM	66.99 +/- 2.71	46.137	90.476
1/8	Random QR	68.02 +/- 2.02	12.583	56.921
1/8	Uncentered SVD	69.47 +/- 2.08	12.583	56.921
1/8	Centered-fit SVD	69.66 +/- 3.91	12.583	56.921
1/8	Learned channel	68.22 +/- 1.22	12.616	56.954
1/4	Learned CM	68.17 +/- 2.73	117.441	161.779
1/4	Random QR	69.47 +/- 2.45	50.332	94.670
1/4	Uncentered SVD	69.34 +/- 1.88	50.332	94.670
1/4	Centered-fit SVD	69.70 +/- 3.46	50.332	94.670
1/4	Learned channel	69.07 +/- 2.52	50.397	94.736

At matched rho, fixed QR and fixed SVD have the same runtime structure and parameter count; QR avoids data-dependent basis fitting.

Computation: distinguish model cost from stopping time

Recorded training time is affected by stopping epochs and shared-GPU load; it is not a controlled single-step speed benchmark.

rho	Group	Mean train GPU-min	Stop epoch range	Peak process MiB
1/16	Uncentered SVD	6.59	15-22	7906
1/16	Centered-fit SVD	8.12	14-24	7906
1/16	Learned channel	4.07	12-18	7906
1/8	Uncentered SVD	5.27	16-20	8012
1/8	Centered-fit SVD	6.86	9-28	8012
1/8	Learned channel	4.06	12-16	8012
1/4	Uncentered SVD	5.84	12-18	8144
1/4	Centered-fit SVD	5.51	13-14	8144
1/4	Learned channel	6.28	17-23	8114

Accepted 129 stage-two trainings sum to 610.97 GPU-min. Inherited project accounting totals 1057.95 GPU-min, including earlier studies and separately recorded preparation, diagnostics and failures.

All accepted stage-two process peaks are below 10 GiB. The learned-channel four-mode preflight peak was 8572 MiB. No GPU-time budget was imposed.

At initialization: subspace retention and residual magnitude

Standard Radon, three-seed mean; all 1,264 training participants. Evaluated at bridge input before residual write-back.

Raw energy retention

	CFP	OCT
U 1/16	0.649	0.662
C 1/16	0.476	0.519
L 1/16	0.062	0.067
U 1/8	0.746	0.755
C 1/8	0.612	0.642
L 1/8	0.129	0.132
U 1/4	0.838	0.844
C 1/4	0.732	0.766
L 1/4	0.244	0.269

Centered variance retention

	CFP	OCT
U 1/16	0.483	0.498
C 1/16	0.491	0.506
L 1/16	0.059	0.062
U 1/8	0.626	0.636
C 1/8	0.629	0.640
L 1/8	0.123	0.124
U 1/4	0.762	0.769
C 1/4	0.763	0.770
L 1/4	0.246	0.249

Residual / input L2

	CFP	OCT
U 1/16	0.000	0.000
C 1/16	0.000	0.000
L 1/16	0.000	0.000
U 1/8	0.000	0.000
C 1/8	0.000	0.000
L 1/8	0.000	0.000
U 1/4	0.000	0.000
C 1/4	0.000	0.000
L 1/4	0.000	0.000

U/C use the actual fixed orthogonal subspace. L uses an orthogonal diagnostic projector onto the current encoder row space; learned encoder amplification is recorded separately. Current-model features differ after training. Energy retention is not task information.

After joint training: subspace retention and residual magnitude

Standard Radon, three-seed mean; all 1,264 training participants. Evaluated at bridge input before residual write-back.

Raw energy retention

	CFP	OCT
U 1/16	0.577	0.521
C 1/16	0.407	0.382
L 1/16	0.044	0.041
U 1/8	0.630	0.597
C 1/8	0.518	0.485
L 1/8	0.094	0.078
U 1/4	0.752	0.732
C 1/4	0.649	0.650
L 1/4	0.214	0.195

Centered variance retention

	CFP	OCT
U 1/16	0.382	0.301
C 1/16	0.392	0.322
L 1/16	0.055	0.052
U 1/8	0.463	0.399
C 1/8	0.491	0.422
L 1/8	0.112	0.099
U 1/4	0.644	0.606
C 1/4	0.640	0.611
L 1/4	0.232	0.215

Residual / input L2

	CFP	OCT
U 1/16	0.576	1.723
C 1/16	0.638	2.186
L 1/16	0.165	0.339
U 1/8	1.867	5.730
C 1/8	2.245	10.091
L 1/8	0.427	1.239
U 1/4	4.773	10.415
C 1/4	8.126	19.817
L 1/4	2.924	8.870

U/C use the actual fixed orthogonal subspace. L uses an orthogonal diagnostic projector onto the current encoder row space; learned encoder amplification is recorded separately. Current-model features differ after training. Energy retention is not task information.

Gradient diagnostics: local structural checks, not global guarantees

Each CE gradient is accumulated with participant weighting over the same 128 training participants in eval mode, then cosine is computed. Table: mean over defined seeds (count/3).

rho	Group	CFP stage3 cosine	OCT stage3 cosine	Mixer cosine
1/16	Uncentered SVD	+0.137 (3/3)	+0.173 (3/3)	+0.000 (3/3)
1/16	Centered-fit SVD	+0.160 (3/3)	+0.221 (3/3)	+0.000 (3/3)
1/16	Learned channel	+0.289 (3/3)	+0.146 (3/3)	+0.000 (3/3)
1/8	Uncentered SVD	+0.270 (3/3)	+0.048 (3/3)	+0.000 (3/3)
1/8	Centered-fit SVD	+0.092 (3/3)	+0.247 (3/3)	+0.000 (3/3)
1/8	Learned channel	-0.195 (3/3)	-0.164 (3/3)	+0.000 (3/3)
1/4	Uncentered SVD	+0.217 (3/3)	+0.411 (3/3)	+0.000 (3/3)
1/4	Centered-fit SVD	+0.846 (3/3)	+0.743 (3/3)	+0.000 (3/3)
1/4	Learned channel	+0.511 (3/3)	+0.478 (3/3)	+0.000 (3/3)

The single-bridge mixer has disjoint destination-row gradient support for the two tasks; zero cosine verifies that structure. It does not prove that native networks or learned channel encoders are free of gradient conflict.

Zero-norm gradients are undefined, never assigned cosine 0. Encoder/decoder gradient norms, cosines and learned spectra are retained in the machine-readable evidence.

Interpretation and what the study does not establish

Use a robust working reference; keep the candidate comparison open rather than selecting the largest single score.

Supported descriptive findings. Uncentered SVD performs well over all tested widths. Centered fitting has larger seed variation at the wider settings. Learned channel compression is trainable end-to-end, but its average standard-Radon performance does not exceed uncentered SVD at any tested width.

Mechanistic evidence is conditional. Several Radon-versus-control contrasts favor the geometric bridge, while other intervals cross zero and some seed-wise controls win. Self-only is not effectively parameter-matched. Fixed-versus-learned comparisons also change constraints and, for SVD, initialization.

No universal best rho, monotonicity or ceiling is established. A larger nested subspace retains at least as much energy for the same fixed features. That fact does not guarantee improved development F1 after separate end-to-end training.

Evidence boundary. The development set informed task/configuration selection. Three training seeds do not create an external validation cohort. The sealed test set was not read. This report supports research decisions, not clinical or independent-test claims.

Provenance and acceptance

Immutable experiment directories and source commits; full configuration, prediction and summary hashes are retained locally.

Stage	Run directory suffix	Training source	Accepted results
Three-seed mechanisms	2026_09_05_09_29_16	359306572437	57
Centered-fit supplement	2026_09_05_10_09_52	e0c49f289d6b	+36 = 93
Learned-channel supplement	2026_09_05_13_32_31	35675f444cf7	+36 = 129

Audits: all 129 results reached validation plateau; all referenced diagnostics passed. The 36 three-way configuration comparisons share seed, parent checkpoints, training controls, M/S and rho. New-mode GPU forward/backward/optimizer and full diagnostic preflights passed before deployment.

Evidence files: results.csv lists all 129 scores, gains, parameters, costs, epochs and hashes; paired_differences.csv lists the learned-channel contrasts. The prior centered and uncentered comparison files remain separate and are included in the evidence package. Full results.json preserves diagnostic and checkpoint provenance.

Next pages list every accepted stage-two result. U = uncentered SVD; C = centered-fit SVD; L = learned channel. F1 values are percentages. Best/stop are epoch numbers; best 0 denotes the initial parent predictions.

All results: seed 3416 / part 1 of 2

U / C / L identify the three main channel families. Reference rows are explicitly named. No failed or capped trial is counted.

rho	Method / mechanism	CFP	OCT	Mean	Best / stop	Bridge M	Effective M	GPU-min	Peak MiB
-	No bridge	66.89	65.14	66.01	0 / 8	0.000	0.000	3.84	7796
1/16	Learned CM	70.22	70.18	70.20	11 / 17	19.923	19.923	8.44	8232
1/16	Random QR	66.89	65.14	66.01	0 / 8	3.146	3.146	2.40	7906
1/16	U / Radon	69.57	69.21	69.39	9 / 15	3.146	3.146	4.51	7906
1/16	U / Self only	66.89	65.14	66.01	0 / 8	3.146	1.573	2.42	7906
1/16	U / Scrambled	68.10	65.71	66.90	7 / 13	3.146	3.146	3.79	7906
1/16	U / Resampling	66.89	65.14	66.01	0 / 8	3.146	3.146	2.32	7848
1/16	C / Radon	67.53	67.88	67.70	8 / 14	3.146	3.146	6.71	7906
1/16	C / Self only	66.89	65.14	66.01	0 / 8	3.146	1.573	2.41	7906
1/16	C / Scrambled	69.26	68.10	68.68	10 / 16	3.146	3.146	4.59	7906
1/16	C / Resampling	66.89	65.14	66.01	0 / 8	3.146	3.146	3.80	7848
1/16	L / Radon	67.42	67.09	67.26	7 / 13	3.162	3.162	3.75	7906
1/16	L / Self only	66.89	65.14	66.01	0 / 8	3.162	1.589	2.43	7906
1/16	L / Scrambled	68.15	65.52	66.83	7 / 13	3.162	3.162	3.77	7906
1/16	L / Resampling	66.89	65.14	66.01	0 / 8	3.162	3.162	2.29	7848
1/8	Learned CM	69.64	70.53	70.09	16 / 22	46.137	46.137	19.05	8662
1/8	Random QR	71.21	69.25	70.23	7 / 13	12.583	12.583	3.95	8012
1/8	U / Radon	72.30	71.28	71.79	11 / 17	12.583	12.583	5.08	8012
1/8	U / Self only	66.89	65.14	66.01	0 / 8	12.583	6.291	2.51	8012
1/8	U / Scrambled	68.24	69.21	68.73	10 / 10	12.583	12.583	3.10	8012
1/8	U / Resampling	69.59	65.46	67.53	9 / 15	12.583	12.583	4.20	8154
1/8	C / Radon	74.61	73.30	73.96	28 / 28	12.583	12.583	10.38	8012

Exact result IDs, full hashes, total native-plus-bridge parameters and baseline gains are available in the accompanying CSV.

All results: seed 3416 / part 2 of 2

U / C / L identify the three main channel families. Reference rows are explicitly named. No failed or capped trial is counted.

rho	Method / mechanism	CFP	OCT	Mean	Best / stop	Bridge M	Effective M	GPU-min	Peak MiB
1/8	C / Self only	66.89	65.14	66.01	0 / 8	12.583	6.291	2.52	8012
1/8	C / Scrambled	70.94	69.18	70.06	8 / 14	12.583	12.583	4.24	8012
1/8	C / Resampling	66.89	65.14	66.01	0 / 8	12.583	12.583	2.37	8154
1/8	L / Radon	70.92	68.24	69.58	10 / 16	12.616	12.616	4.72	8012
1/8	L / Self only	66.89	65.14	66.01	0 / 8	12.616	6.324	2.51	8012
1/8	L / Scrambled	71.61	67.77	69.69	10 / 16	12.616	12.616	4.80	8012
1/8	L / Resampling	66.89	65.14	66.01	0 / 8	12.616	12.616	2.34	8154
1/4	Learned CM	69.90	70.20	70.05	10 / 16	117.441	117.441	9.18	9552
1/4	Random QR	72.25	72.28	72.26	9 / 15	50.332	50.332	5.10	8144
1/4	U / Radon	72.97	69.56	71.27	9 / 15	50.332	50.332	7.29	8144
1/4	U / Self only	66.89	65.14	66.01	0 / 8	50.332	25.166	4.69	8144
1/4	U / Scrambled	68.48	70.51	69.49	14 / 20	50.332	50.332	6.66	8144
1/4	U / Resampling	69.20	66.19	67.69	11 / 17	50.332	50.332	8.82	8096
1/4	C / Radon	73.59	73.53	73.56	7 / 13	50.332	50.332	7.39	8144
1/4	C / Self only	66.89	65.14	66.01	0 / 8	50.332	25.166	2.84	8144
1/4	C / Scrambled	73.59	70.59	72.09	7 / 13	50.332	50.332	4.45	8144
1/4	C / Resampling	69.57	67.82	68.70	5 / 11	50.332	50.332	3.49	8096
1/4	L / Radon	71.27	72.62	71.94	11 / 17	50.397	50.397	5.65	8114
1/4	L / Self only	66.89	65.14	66.01	0 / 8	50.397	25.231	2.91	8114
1/4	L / Scrambled	68.17	70.16	69.17	9 / 13	50.397	50.397	4.44	8114
1/4	L / Resampling	69.58	67.90	68.74	15 / 21	50.397	50.397	6.29	8132

Exact result IDs, full hashes, total native-plus-bridge parameters and baseline gains are available in the accompanying CSV.

All results: seed 3417 / part 1 of 2

U / C / L identify the three main channel families. Reference rows are explicitly named. No failed or capped trial is counted.

rho	Method / mechanism	CFP	OCT	Mean	Best / stop	Bridge M	Effective M	GPU-min	Peak MiB
-	No bridge	63.85	64.19	64.02	0 / 8	0.000	0.000	3.77	7796
1/16	Learned CM	66.89	65.77	66.33	7 / 13	19.923	19.923	6.46	8232
1/16	Random QR	63.85	64.19	64.02	0 / 8	3.146	3.146	2.44	7906
1/16	U / Radon	68.19	68.56	68.38	16 / 22	3.146	3.146	9.51	7906
1/16	U / Self only	63.85	64.19	64.02	0 / 8	3.146	1.573	4.04	7906
1/16	U / Scrambled	67.89	63.50	65.69	11 / 17	3.146	3.146	4.87	7906
1/16	U / Resampling	63.85	64.19	64.02	0 / 8	3.146	3.146	3.77	7848
1/16	C / Radon	68.75	66.87	67.81	18 / 24	3.146	3.146	11.15	7906
1/16	C / Self only	63.85	64.19	64.02	0 / 8	3.146	1.573	2.46	7906
1/16	C / Scrambled	64.53	67.57	66.05	11 / 17	3.146	3.146	4.91	7906
1/16	C / Resampling	63.85	64.19	64.02	0 / 8	3.146	3.146	2.28	7848
1/16	L / Radon	65.79	63.34	64.56	6 / 12	3.162	3.162	3.43	7906
1/16	L / Self only	63.85	64.19	64.02	0 / 8	3.162	1.589	2.39	7906
1/16	L / Scrambled	63.85	64.19	64.02	0 / 8	3.162	3.162	2.41	7906
1/16	L / Resampling	63.85	64.19	64.02	0 / 8	3.162	3.162	2.28	7848
1/8	Learned CM	65.88	65.83	65.85	10 / 13	46.137	46.137	11.38	8662
1/8	Random QR	69.07	63.47	66.27	6 / 12	12.583	12.583	3.70	8012
1/8	U / Radon	69.83	67.89	68.86	10 / 16	12.583	12.583	4.80	8012
1/8	U / Self only	63.85	64.19	64.02	0 / 8	12.583	6.291	4.20	8012
1/8	U / Scrambled	65.54	67.91	66.72	10 / 16	12.583	12.583	4.80	8012
1/8	U / Resampling	63.85	64.19	64.02	0 / 8	12.583	12.583	4.18	8154
1/8	C / Radon	68.87	68.57	68.72	9 / 15	12.583	12.583	7.40	8012

Exact result IDs, full hashes, total native-plus-bridge parameters and baseline gains are available in the accompanying CSV.

All results: seed 3417 / part 2 of 2

U / C / L identify the three main channel families. Reference rows are explicitly named. No failed or capped trial is counted.

rho	Method / mechanism	CFP	OCT	Mean	Best / stop	Bridge M	Effective M	GPU-min	Peak MiB
1/8	C / Self only	63.85	64.19	64.02	0 / 8	12.583	6.291	2.55	8012
1/8	C / Scrambled	67.57	68.55	68.06	11 / 17	12.583	12.583	5.08	8012
1/8	C / Resampling	64.44	64.02	64.23	6 / 12	12.583	12.583	4.16	8154
1/8	L / Radon	67.47	68.21	67.84	6 / 12	12.616	12.616	3.59	8012
1/8	L / Self only	63.85	64.19	64.02	0 / 8	12.616	6.324	2.51	8012
1/8	L / Scrambled	65.54	63.84	64.69	10 / 16	12.616	12.616	4.76	8012
1/8	L / Resampling	63.85	64.19	64.02	0 / 8	12.616	12.616	2.34	8154
1/4	Learned CM	68.91	69.93	69.42	10 / 16	117.441	117.441	9.20	9552
1/4	Random QR	68.56	68.24	68.40	10 / 16	50.332	50.332	5.42	8144
1/4	U / Radon	69.92	68.57	69.24	12 / 18	50.332	50.332	6.09	8144
1/4	U / Self only	63.85	64.19	64.02	0 / 8	50.332	25.166	4.78	8144
1/4	U / Scrambled	68.17	67.47	67.82	11 / 17	50.332	50.332	5.76	8144
1/4	U / Resampling	65.38	64.02	64.70	6 / 12	50.332	50.332	5.69	8096
1/4	C / Radon	67.77	69.57	68.67	8 / 14	50.332	50.332	4.74	8144
1/4	C / Self only	63.85	64.19	64.02	0 / 8	50.332	25.166	4.24	8144
1/4	C / Scrambled	68.91	62.30	65.61	6 / 12	50.332	50.332	8.36	8144
1/4	C / Resampling	67.18	63.17	65.18	8 / 14	50.332	50.332	8.85	8096
1/4	L / Radon	68.58	67.54	68.06	11 / 17	50.397	50.397	5.66	8114
1/4	L / Self only	63.85	64.19	64.02	0 / 8	50.397	25.231	2.81	8114
1/4	L / Scrambled	69.13	65.17	67.15	8 / 14	50.397	50.397	4.80	8114
1/4	L / Resampling	63.85	64.19	64.02	0 / 8	50.397	50.397	2.58	8132

Exact result IDs, full hashes, total native-plus-bridge parameters and baseline gains are available in the accompanying CSV.

All results: seed 3418 / part 1 of 2

U / C / L identify the three main channel families. Reference rows are explicitly named. No failed or capped trial is counted.

rho	Method / mechanism	CFP	OCT	Mean	Best / stop	Bridge M	Effective M	GPU-min	Peak MiB
-	No bridge	65.88	64.19	65.03	0 / 8	0.000	0.000	3.65	7796
1/16	Learned CM	68.48	68.10	68.29	8 / 14	19.923	19.923	11.85	8232
1/16	Random QR	65.53	67.23	66.38	9 / 15	3.146	3.146	6.97	7906
1/16	U / Radon	68.92	68.55	68.73	14 / 20	3.146	3.146	5.76	7906
1/16	U / Self only	65.88	64.19	65.03	0 / 8	3.146	1.573	2.44	7906
1/16	U / Scrambled	65.81	65.95	65.88	5 / 11	3.146	3.146	3.24	7906
1/16	U / Resampling	66.84	64.17	65.51	10 / 14	3.146	3.146	3.84	7848
1/16	C / Radon	68.55	67.84	68.20	17 / 23	3.146	3.146	6.49	7906
1/16	C / Self only	65.88	64.19	65.03	0 / 8	3.146	1.573	2.46	7906
1/16	C / Scrambled	68.91	65.87	67.39	13 / 19	3.146	3.146	5.44	7906
1/16	C / Resampling	65.88	64.19	65.03	0 / 8	3.146	3.146	2.30	7848
1/16	L / Radon	66.89	68.23	67.56	12 / 18	3.162	3.162	5.02	7906
1/16	L / Self only	65.88	64.19	65.03	0 / 8	3.162	1.589	2.39	7906
1/16	L / Scrambled	65.88	64.19	65.03	0 / 8	3.162	3.162	2.39	7906
1/16	L / Resampling	65.88	64.19	65.03	0 / 8	3.162	3.162	2.28	7848
1/8	Learned CM	65.88	64.19	65.03	0 / 8	46.137	46.137	7.05	8662
1/8	Random QR	67.67	67.42	67.55	14 / 20	12.583	12.583	5.96	8012
1/8	U / Radon	68.10	67.42	67.76	14 / 20	12.583	12.583	5.93	8012
1/8	U / Self only	65.88	64.19	65.03	0 / 8	12.583	6.291	2.53	8012
1/8	U / Scrambled	65.20	68.19	66.70	9 / 13	12.583	12.583	6.48	8012
1/8	U / Resampling	65.88	65.38	65.63	12 / 12	12.583	12.583	3.44	8154
1/8	C / Radon	67.82	64.79	66.30	3 / 9	12.583	12.583	2.80	8012

Exact result IDs, full hashes, total native-plus-bridge parameters and baseline gains are available in the accompanying CSV.

All results: seed 3418 / part 2 of 2

U / C / L identify the three main channel families. Reference rows are explicitly named. No failed or capped trial is counted.

rho	Method / mechanism	CFP	OCT	Mean	Best / stop	Bridge M	Effective M	GPU-min	Peak MiB
1/8	C / Self only	65.88	64.19	65.03	0 / 8	12.583	6.291	2.53	8012
1/8	C / Scrambled	65.03	66.79	65.91	12 / 18	12.583	12.583	5.35	8012
1/8	C / Resampling	65.88	64.19	65.03	0 / 8	12.583	12.583	2.34	8154
1/8	L / Radon	66.55	67.91	67.23	7 / 13	12.616	12.616	3.87	8012
1/8	L / Self only	65.88	64.19	65.03	0 / 8	12.616	6.324	2.51	8012
1/8	L / Scrambled	67.57	66.55	67.06	10 / 16	12.616	12.616	4.76	8012
1/8	L / Resampling	68.54	64.17	66.36	7 / 13	12.616	12.616	3.63	8154
1/4	Learned CM	65.88	64.19	65.03	0 / 8	117.441	117.441	5.40	9082
1/4	Random QR	67.56	67.90	67.73	12 / 18	50.332	50.332	6.07	8144
1/4	U / Radon	68.22	66.82	67.52	6 / 12	50.332	50.332	4.13	8144
1/4	U / Self only	65.88	64.19	65.03	0 / 8	50.332	25.166	2.85	8144
1/4	U / Scrambled	67.57	66.88	67.22	9 / 15	50.332	50.332	8.52	8144
1/4	U / Resampling	67.84	69.28	68.56	5 / 11	50.332	50.332	3.52	8096
1/4	C / Radon	67.89	65.85	66.87	7 / 13	50.332	50.332	4.39	8144
1/4	C / Self only	65.88	64.19	65.03	0 / 8	50.332	25.166	2.83	8144
1/4	C / Scrambled	69.54	67.82	68.68	6 / 12	50.332	50.332	4.11	8144
1/4	C / Resampling	65.83	67.77	66.80	8 / 14	50.332	50.332	4.30	8096
1/4	L / Radon	68.88	65.54	67.21	17 / 23	50.397	50.397	7.54	8114
1/4	L / Self only	62.75	69.25	66.00	4 / 10	50.397	25.231	3.49	8114
1/4	L / Scrambled	68.58	67.86	68.22	4 / 10	50.397	50.397	3.49	8114
1/4	L / Resampling	66.89	67.55	67.22	13 / 18	50.397	50.397	5.53	8132

Exact result IDs, full hashes, total native-plus-bridge parameters and baseline gains are available in the accompanying CSV.